

Data Mining Cluster Analysis: Basic Concepts and Algorithms

Lecture Notes for Chapter 7

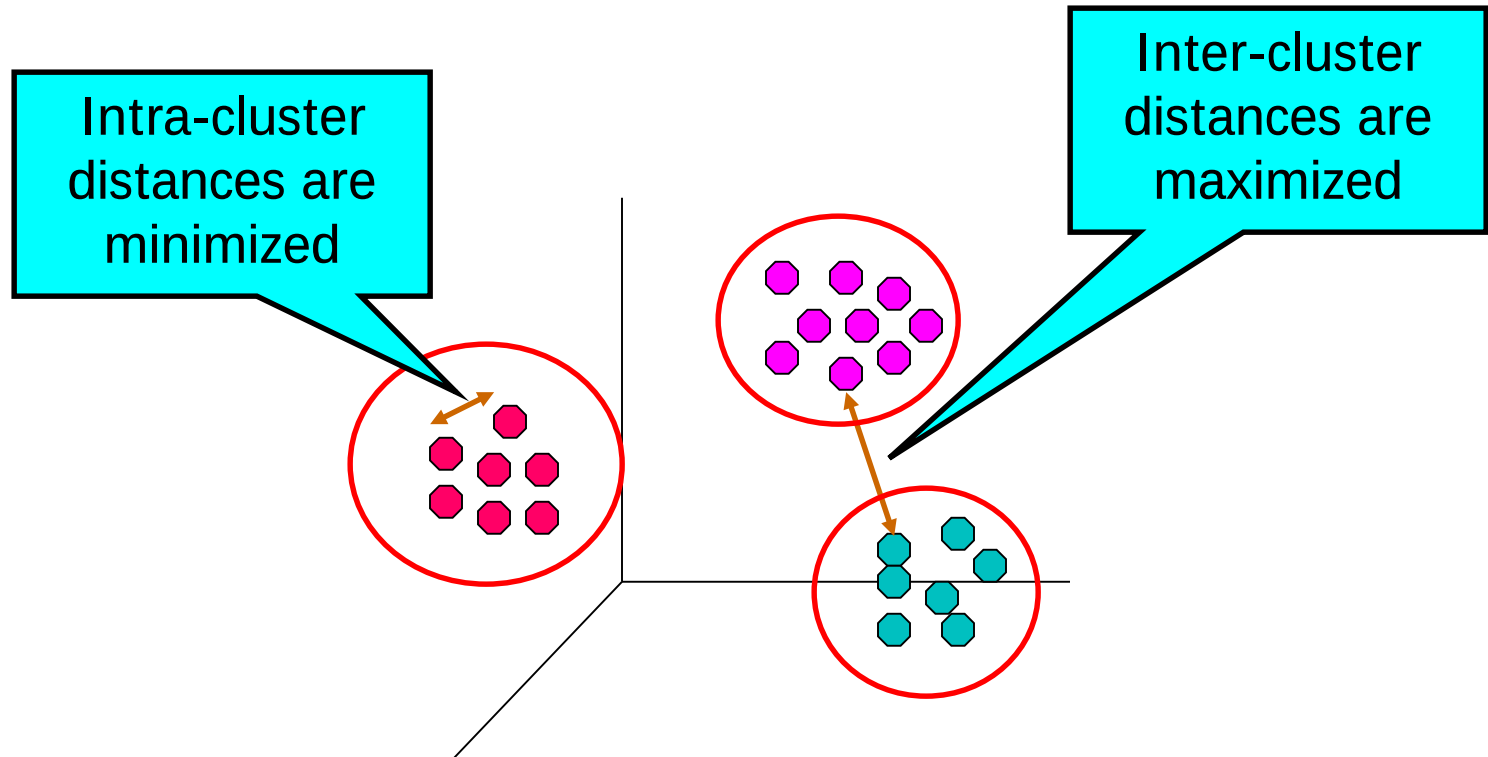
Introduction to Data Mining
by
Tan, Steinbach, Kumar

Outline

- **Introduction**
- K-means
- Agglomerative hierarchical clustering
- DBSCAN
- Cluster evaluation

What is Cluster Analysis?

- Given a set of objects, place them in groups such that the objects in a group are similar (or related) to one another and different from (or unrelated to) the objects in other groups



Applications of Cluster Analysis

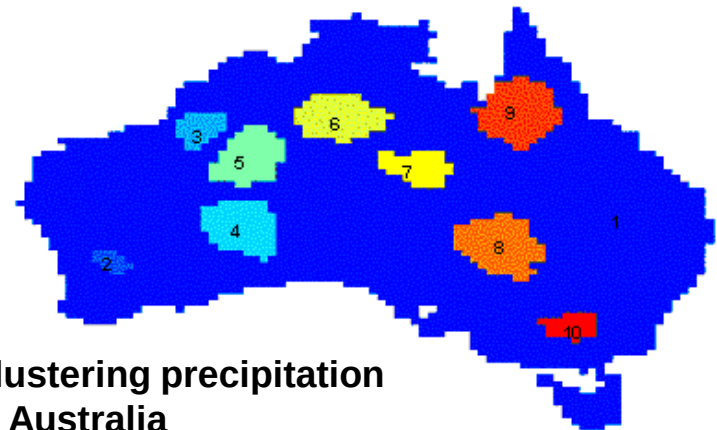
● Understanding

- Group related documents for browsing, group genes and proteins that have similar functionality, or group stocks with similar price fluctuations

	<i>Discovered Clusters</i>	<i>Industry Group</i>
1	Applied-Matl-DOWN,Bay-Network-Down,3-COM-DOWN,Cabletron-Sys-DOWN,CISCO-DOWN,HP-DOWN,DSC-Comm-DOWN,INTEL-DOWN,LSI-Logic-DOWN,Micron-Tech-DOWN,Texas-Inst-Down,Tellabs-Inc-Down,Natl-Semiconduct-DOWN,Oracl-DOWN,SGI-DOWN,Sun-DOWN	Technology1-DOWN
2	Apple-Comp-DOWN,Autodesk-DOWN,DEC-DOWN,ADV-Micro-Device-DOWN,Andrew-Corp-DOWN,Computer-Assoc-DOWN,Circuit-City-DOWN,Compaq-DOWN,EMC-Corp-DOWN,Gen-Inst-DOWN,Motorola-DOWN,Microsoft-DOWN,Scientific-Atl-DOWN	Technology2-DOWN
3	Fannie-Mae-DOWN,Fed-Home-Loan-DOWN,MBNA-Corp-DOWN,Morgan-Stanley-DOWN	Financial-DOWN
4	Baker-Hughes-UP,Dresser-Inds-UP,Halliburton-HLD-UP,Louisiana-Land-UP,Phillips-Petro-UP,Unocal-UP,Schlumberger-UP	Oil-UP

● Summarization

- Reduce the size of large data sets

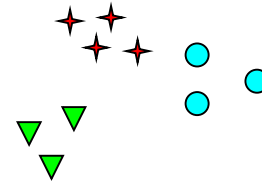


Clustering precipitation
in Australia

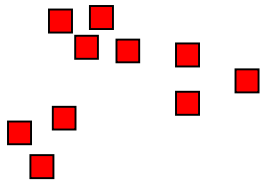
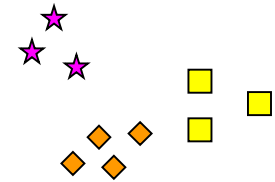
Notion of a Cluster can be Ambiguous



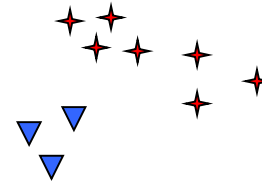
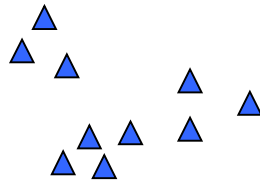
How many clusters?



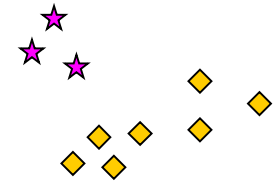
Six Clusters



Two Clusters



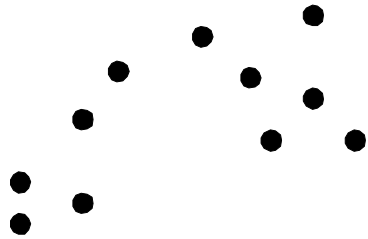
Four Clusters



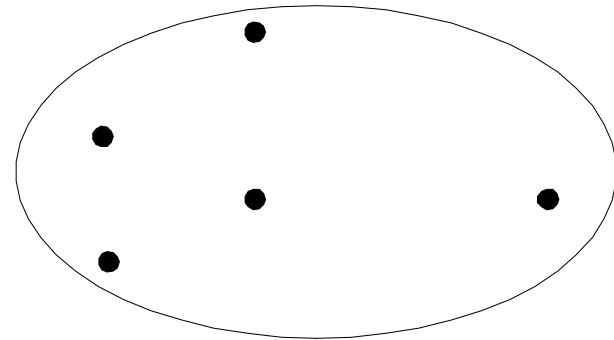
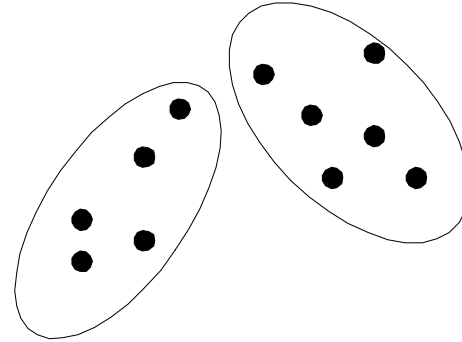
Types of Clusterings

- A **clustering** is a set of clusters
- Important distinction between **hierarchical** and **partitional** sets of clusters
 - Partitional Clustering
 - ◆ A division of data objects into non-overlapping subsets (clusters)
 - Hierarchical clustering
 - ◆ A set of nested clusters organized as a hierarchical tree

Partitional Clustering

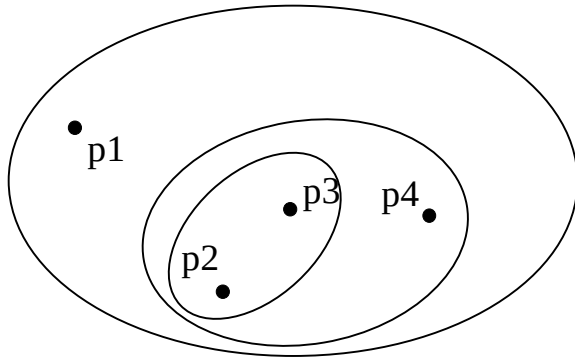


Original Points

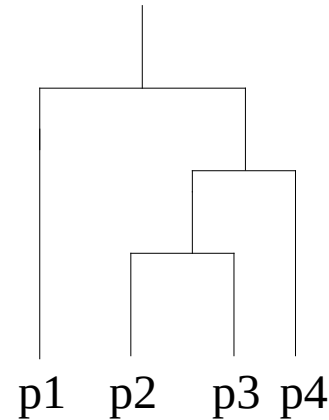


A Partitional Clustering

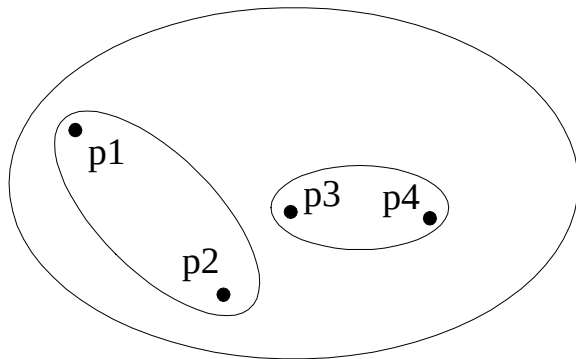
Hierarchical Clustering



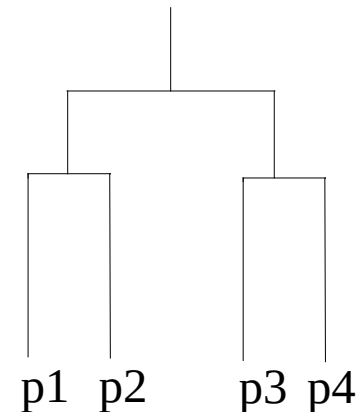
Traditional Hierarchical Clustering



Traditional Dendrogram



Non-traditional Hierarchical Clustering



Non-traditional Dendrogram

Other Distinctions Between Sets of Clusters

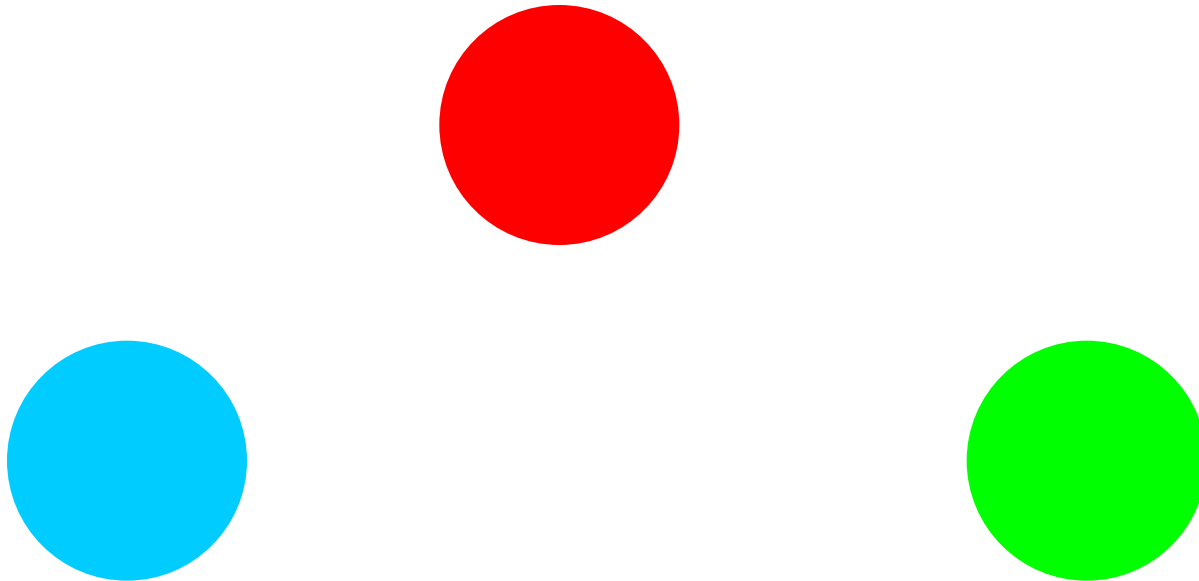
- Exclusive versus non-exclusive
 - In non-exclusive clustering, points may belong to multiple clusters.
 - ◆ Can belong to multiple classes or could be ‘border’ points
 - Fuzzy clustering (one type of non-exclusive)
 - ◆ In fuzzy clustering, a point belongs to every cluster with some weight between 0 and 1
 - ◆ Weights must sum to 1
 - ◆ Probabilistic clustering has similar characteristics
- Partial versus complete
 - In some cases, we only want to cluster some of the data
 - e.g., noise, outliers, or “uninteresting background.”

Types of Clusters

- Well-separated clusters
- Prototype-based clusters
- Contiguity-based clusters
- Density-based clusters
- Described by an Objective Function

Types of Clusters: Well-Separated

- Well-Separated Clusters:
 - A cluster is a set of points such that **any point** in a cluster is closer (or more similar) to every other point in the cluster than to any point not in the cluster.

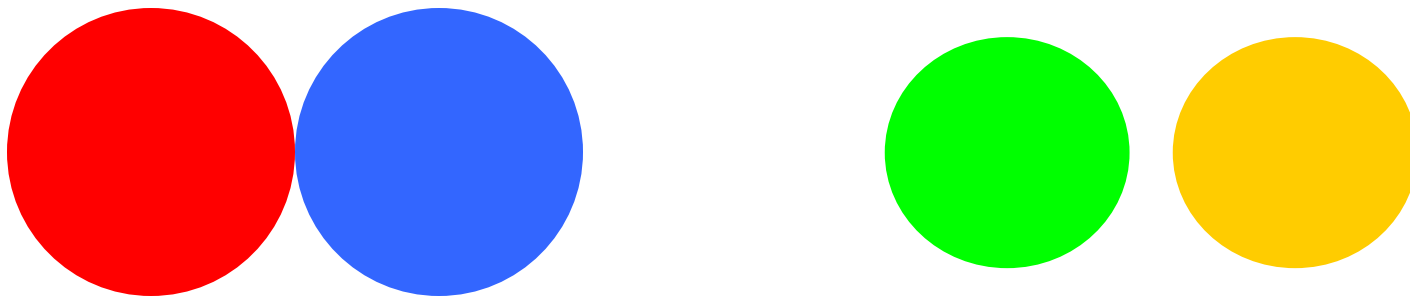


3 well-separated clusters

Types of Clusters: Prototype-Based

- Prototype-based

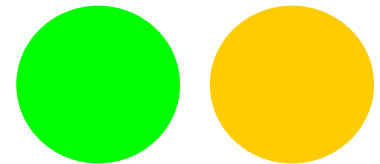
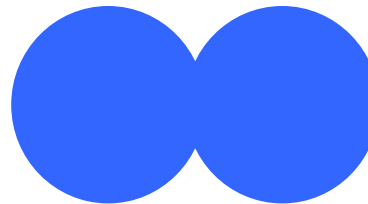
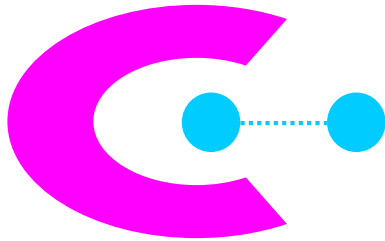
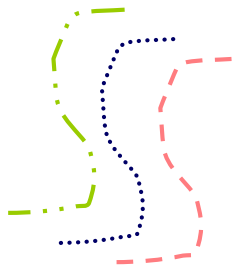
- A cluster is a set of objects such that an object in a cluster is closer (more similar) to **the prototype or “center”** of a cluster, than to the center of any other cluster
- The center of a cluster is often a **centroid**, the average of all the points in the cluster, or a **medoid**, the most “representative” point of a cluster



4 center-based clusters

Types of Clusters: Contiguity-Based / graph-based

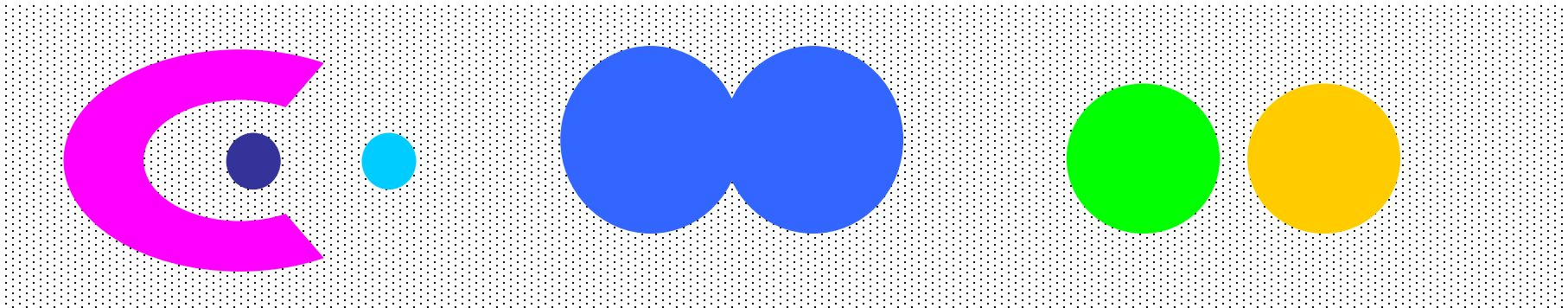
- Contiguous Cluster (Nearest neighbor or Transitive)
 - A cluster is a set of points such that a point in a cluster is closer (or more similar) to **one or more other points in the cluster than to any point not in the cluster.**



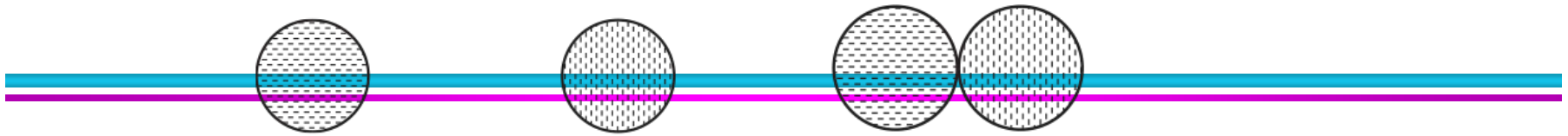
8 contiguous clusters

Types of Clusters: Density-Based

- Density-based
 - A cluster is a dense region of points, which is separated by low-density regions, from other regions of high density.
 - Used when the clusters are irregular or intertwined, and when noise and outliers are present.

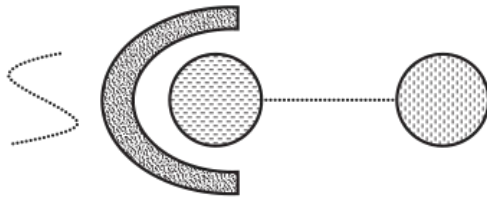


6 density-based clusters

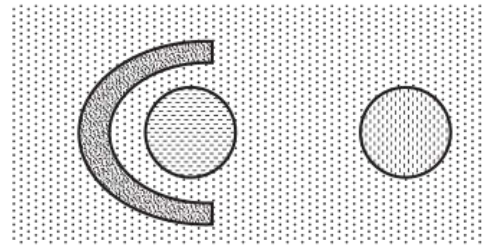


(a) Well-separated clusters. Each point is closer to all of the points in its cluster than to any point in another cluster.

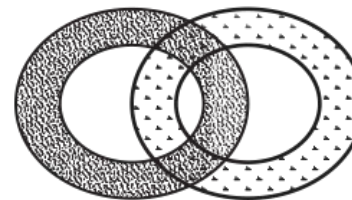
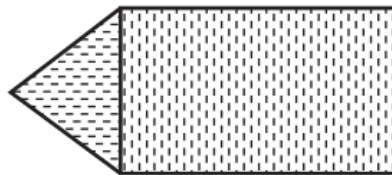
(b) Center-based clusters. Each point is closer to the center of its cluster than to the center of any other cluster.



(c) Contiguity-based clusters. Each point is closer to at least one point in its cluster than to any point in another cluster.



(d) Density-based clusters. Clusters are regions of high density separated by regions of low density.



(e) Conceptual clusters. Points in a cluster share some general property that derives from the entire set of points. (Points in the intersection of the circles belong to both.)

Characteristics of the Input Data Are Important

- Type of proximity or density measure
 - Central to clustering
 - Depends on data and application
- Data characteristics that affect proximity and/or density are
 - Dimensionality
 - ◆ Sparseness
 - Attribute type
 - Special relationships in the data
 - ◆ For example, autocorrelation
 - Distribution of the data
- Noise and Outliers
 - Often interfere with the operation of the clustering algorithm
- Clusters of differing sizes, densities, and shapes

Outline

- Introduction
- **K-means**
 - **Basic algorithm**
 - Choosing Initial Centroids
 - Choosing K
- Agglomerative hierarchical clustering
- DBSCAN
- Cluster evaluation

Clustering Algorithms

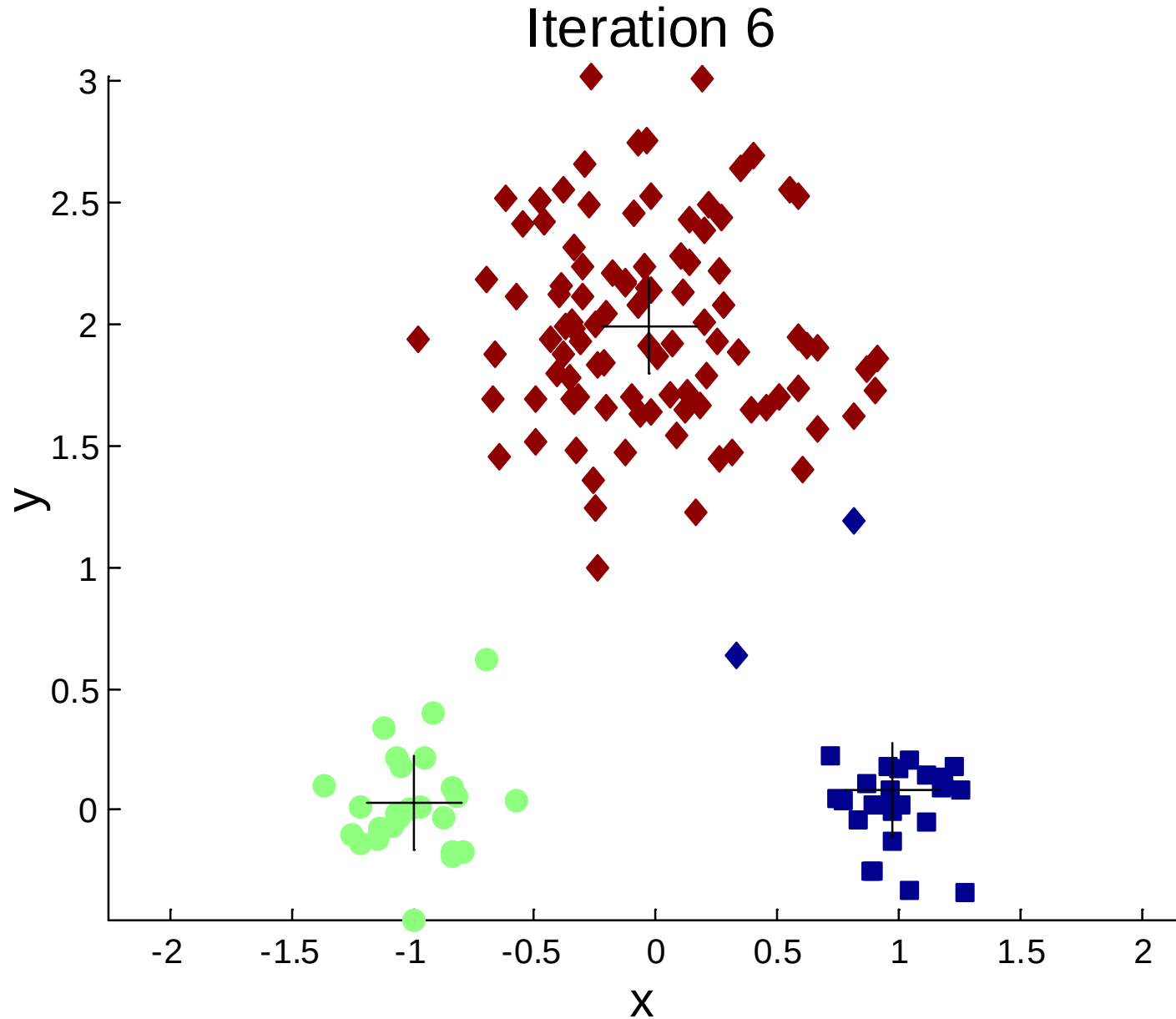
- K-means and its variants
- Hierarchical clustering
- Density-based clustering

K-means Clustering

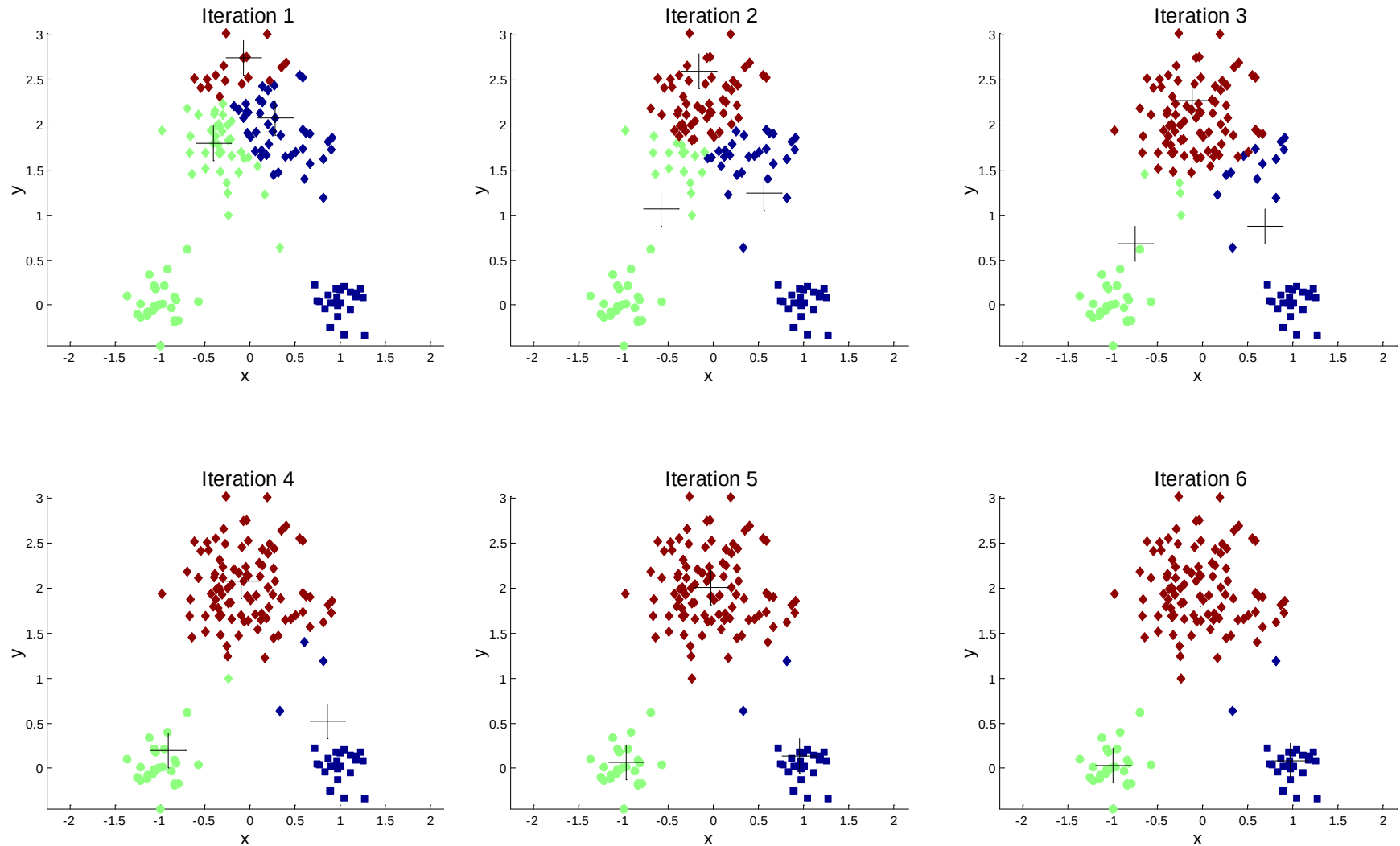
- Partitional clustering approach
- Number of clusters, K , must be specified
- Each cluster is associated with a **centroid** (center point)
- Each point is assigned to the cluster with the closest centroid
- The basic algorithm is very simple

-
- 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change
-

Example of K-means Clustering



Example of K-means Clustering



K-means Clustering – Details

- Simple iterative algorithm.
 - Choose initial centroids;
 - repeat {assign each point to a nearest centroid; re-compute cluster centroids}
 - until centroids stop changing.
- The centroid is (typically) the mean of the points in the cluster, but other definitions are possible.
- Complexity is $O(n * K * I * d)$
 - n = number of points, K = number of clusters,
 I = number of iterations, d = number of attributes

K-means Objective Function

- A common objective function (used with Euclidean distance measure) is Sum of Squared Error (SSE)
 - For each point, the error is the distance to the nearest cluster center
 - To get SSE, we square these errors and sum them.

$$SSE = \sum_{i=1}^K \sum_{x \in C_i} dist^2(m_i, x)$$

- x is a data point in cluster C_i and m_i is the centroid (mean) for cluster C_i
- SSE improves in each iteration of K-means until it reaches a local or global minima.

K-means Objective Function

- Document Data:
 - represented as a document-term matrix as described on The Sparse Data Matrix.
 - Our objective is to maximize the similarity of the documents in a cluster to the cluster centroid; this quantity is known as the cohesion of the cluster.

$$\text{Total Cohesion} = \sum_{i=1}^K \sum_{\mathbf{x} \in C_i} \text{cosine}(\mathbf{x}, \mathbf{c}_i)$$

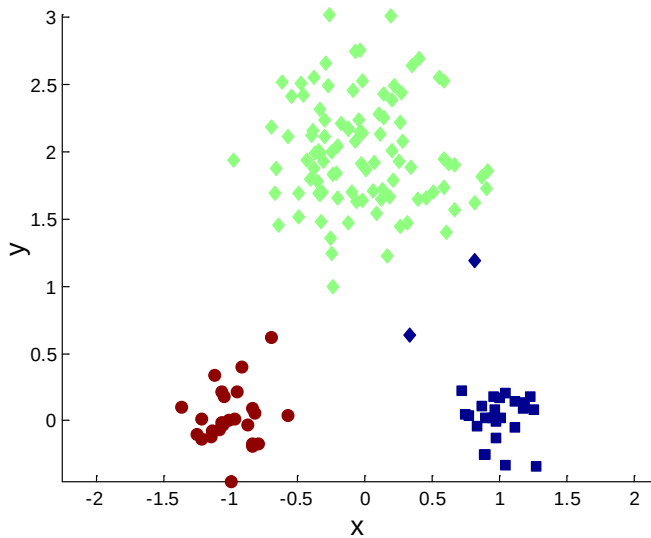
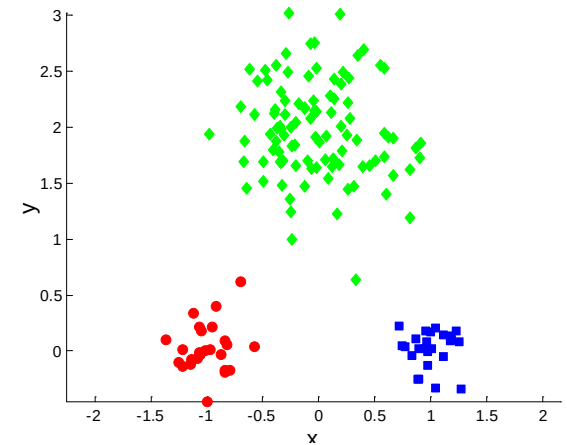
Outline

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- K-means
 - Basic algorithm
 - **Choosing Initial Centroids**
 - Choosing K
- Agglomerative hierarchical clustering
- DBSCAN
- Cluster evaluation

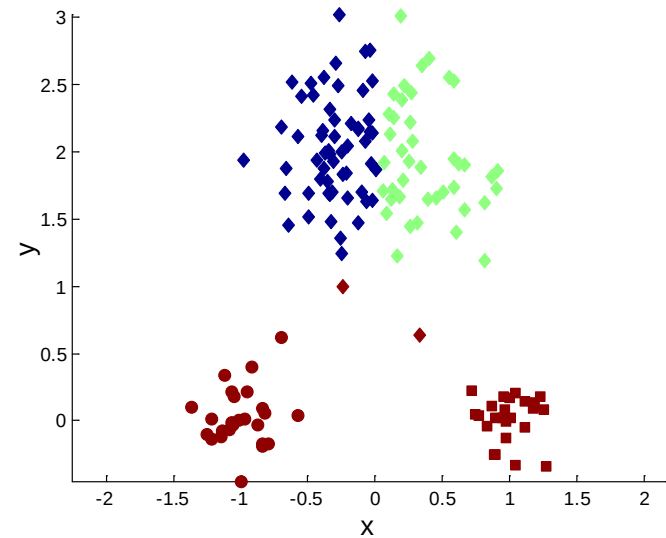
Two different K-means Clusterings

Initial centroids are often chosen randomly.

Clusters produced can vary from one run to another

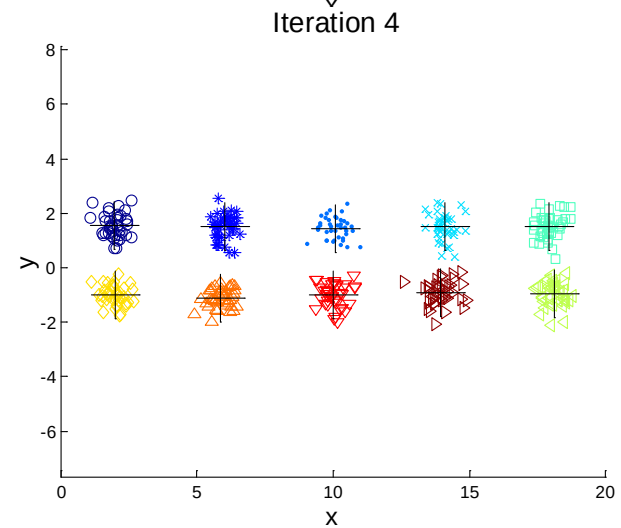
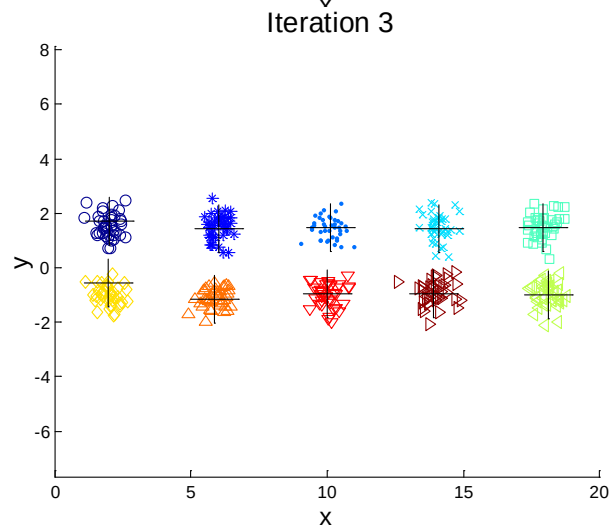
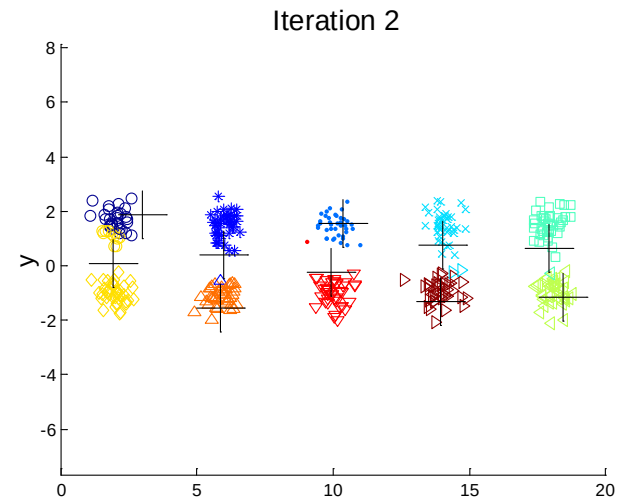
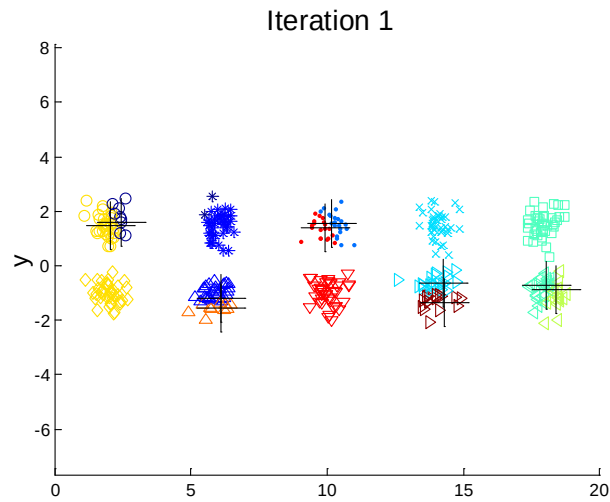


Optimal Clustering



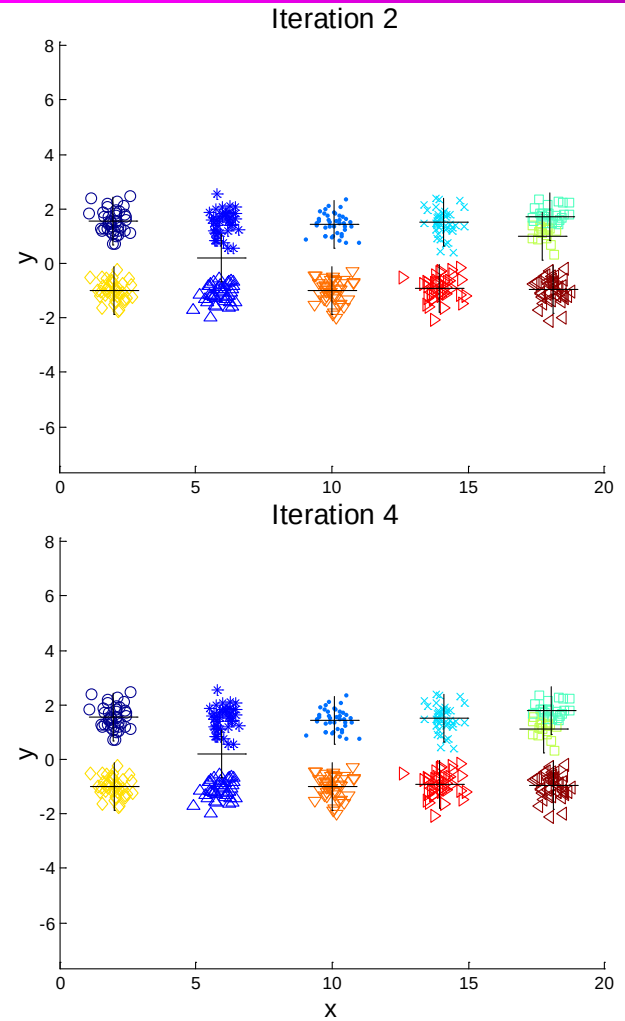
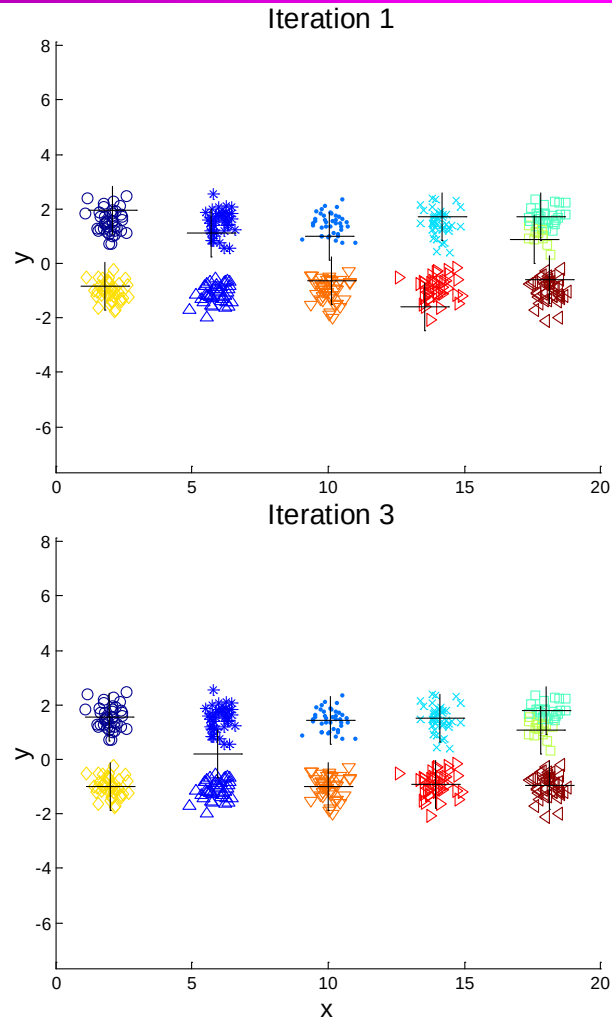
Sub-optimal Clustering

10 Clusters Example



Starting with **two initial centroids in one cluster of each pair of clusters**

10 Clusters Example



Starting with some pairs of clusters having three initial centroids, while other have only one.

Solutions to Initial Centroids Problem

- Multiple runs
 - Helps, but probability is not on your side
- Use some strategy to select the k initial centroids and then select among these initial centroids
 - Select most widely separated
 - ◆ K-means++ is a robust way of doing this selection
 - Use hierarchical clustering to determine initial centroids
- Bisecting K-means
 - Not as susceptible to initialization issues

```
from sklearn.cluster import KMeans

# Initialize KMeans with k-means++ (default behavior)
kmeans = KMeans(n_clusters=3, init='k-means++', random_state=42)
kmeans.fit(X) # Where X is your dataset
```

K-means++

- This approach can be slower than random initialization, but very consistently produces better results in terms of SSE
- To select a set of initial centroids, C , perform the following

- 1: For the first centroid, pick one of the points at random.
- 2: **for** $i = 1$ to *number of trials* **do**
- 3: Compute the distance, $d(x)$, of each point to its closest centroid.
- 4: Assign each point a probability proportional to each point's $d(x)^2$.
- 5: Pick new centroid from the remaining points using the weighted probabilities.
- 6: **end for**

Bisecting K-means

- Bisecting K-means algorithm
 - Variant of K-means that can produce a partitional or a hierarchical clustering

```
1: Initialize the list of clusters to contain the cluster containing all points.  
2: repeat  
3:   Select a cluster from the list of clusters  
4:   for  $i = 1$  to number_of_iterations do  
5:     Bisect the selected cluster using basic K-means  
6:   end for  
7:   Add the two clusters from the bisection with the lowest SSE to the list of clusters.  
8: until Until the list of clusters contains  $K$  clusters
```

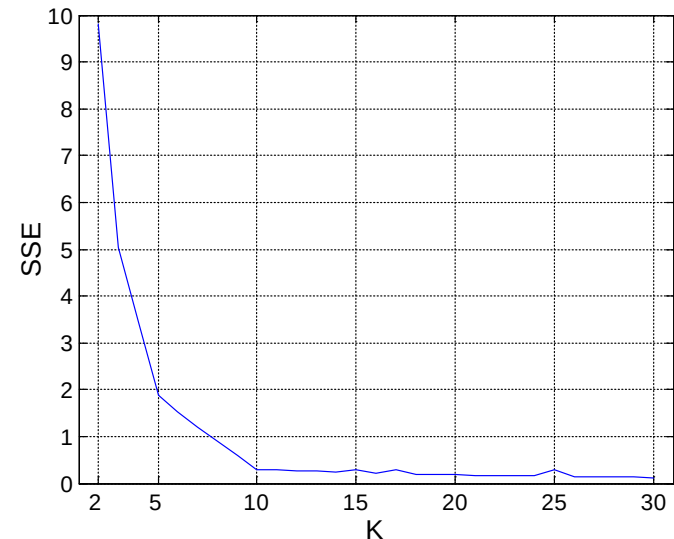
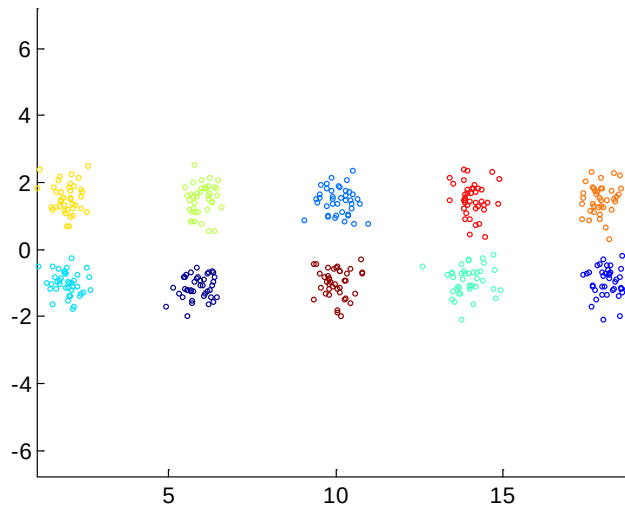
CLUTO: <http://glaros.dtc.umn.edu/gkhome/cluto/cluto/overview>

Outline

- Introduction
- K-means
 - Basic algorithm
 - Choosing Initial Centroids
 - **Choosing K (Elbow method)**
- Agglomerative hierarchical clustering
- DBSCAN
- Cluster evaluation

Determining the Correct Number of Clusters

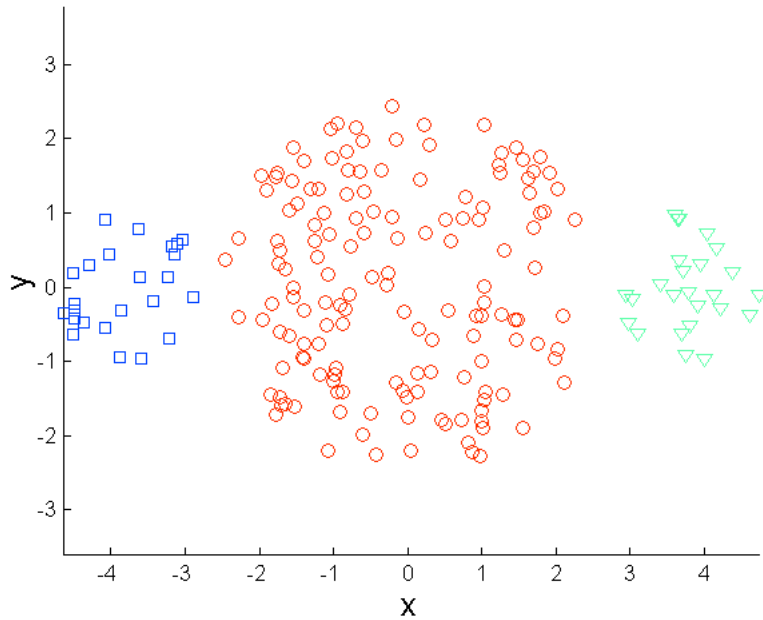
- SSE is good for comparing two clusterings or two clusters
- SSE can also be used to estimate the number of clusters



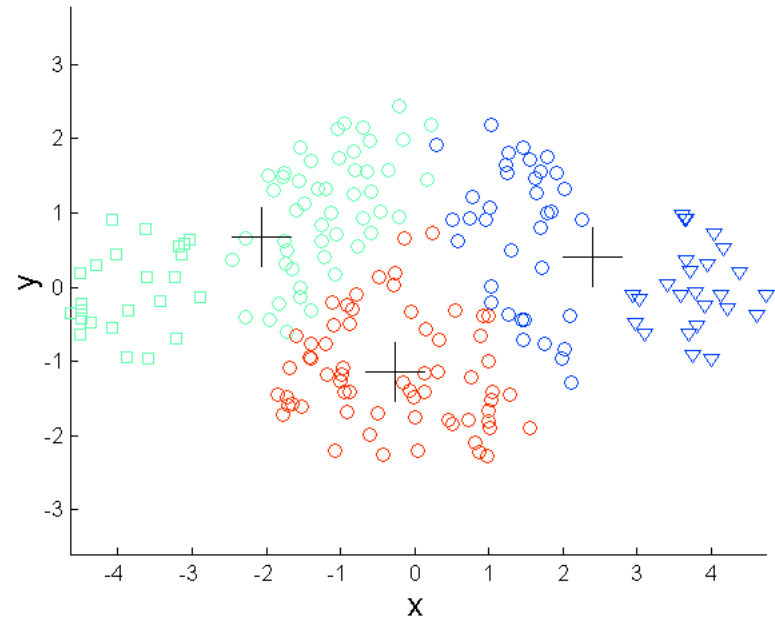
Limitations of K-means

- K-means has problems when clusters are of differing
 - Sizes
 - Densities
 - Non-globular shapes
- K-means has problems when the data contains outliers.
 - One possible solution is to remove outliers before clustering

Limitations of K-means: Differing Sizes

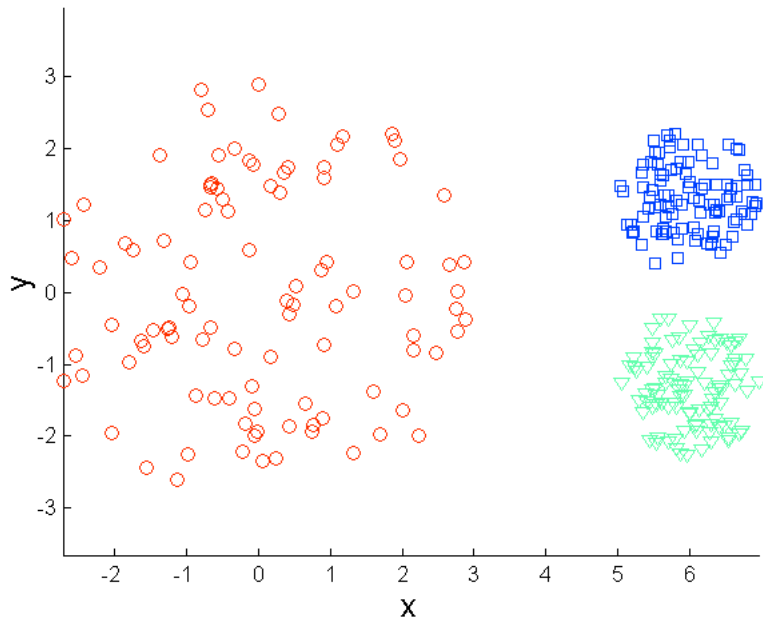


Original Points

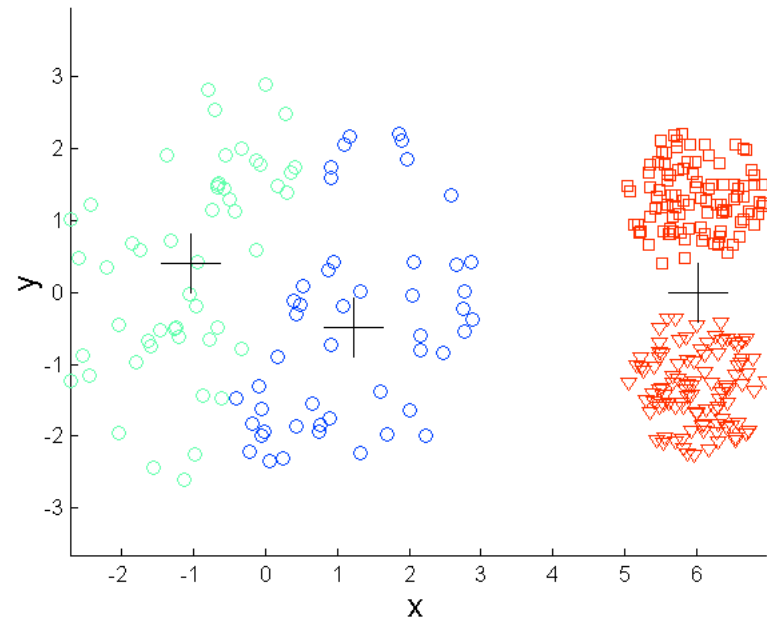


K-means (3 Clusters)

Limitations of K-means: Differing Density

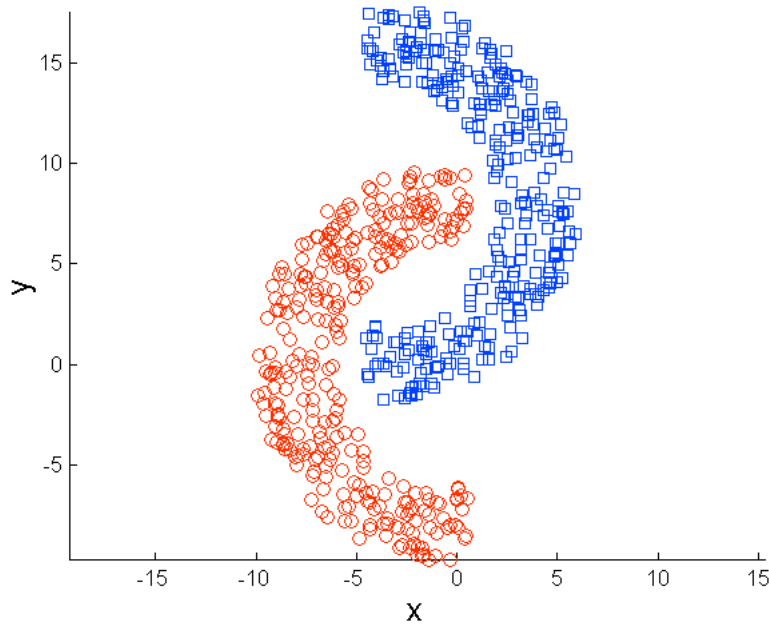


Original Points

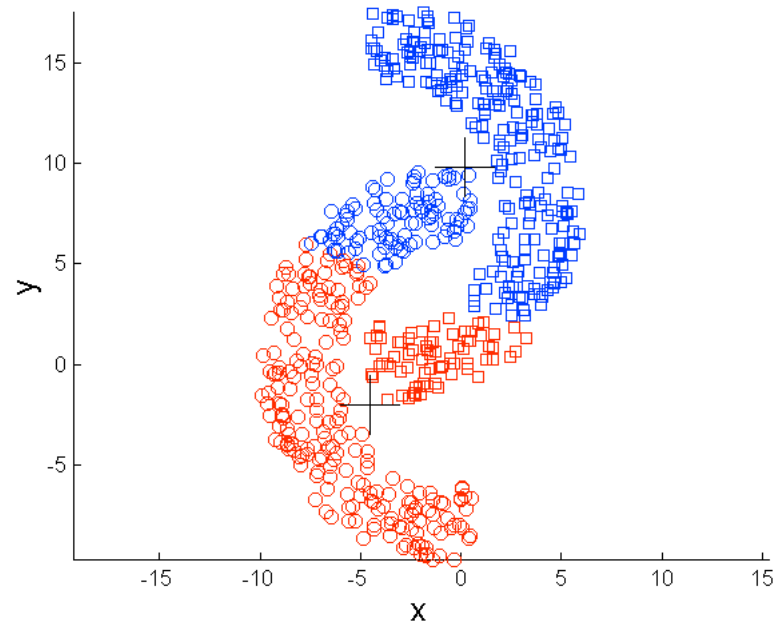


K-means (3 Clusters)

Limitations of K-means: Non-globular Shapes

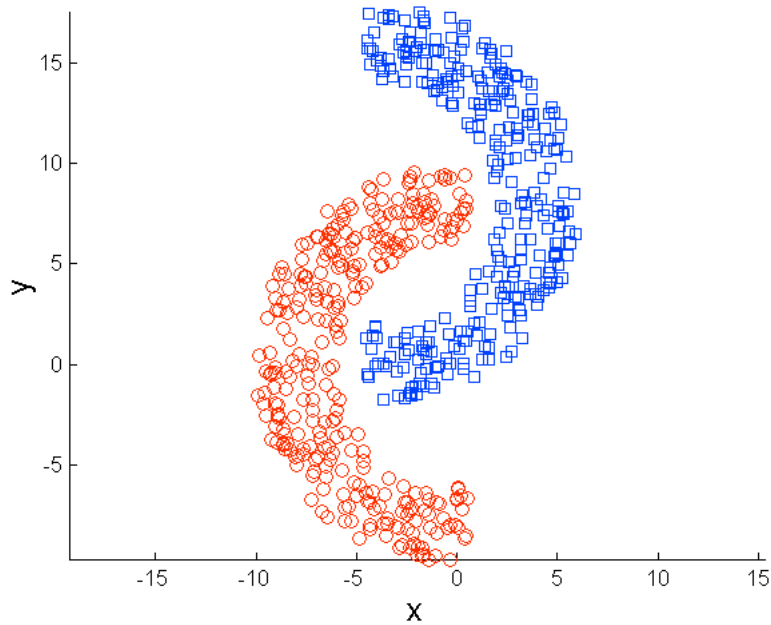


Original Points

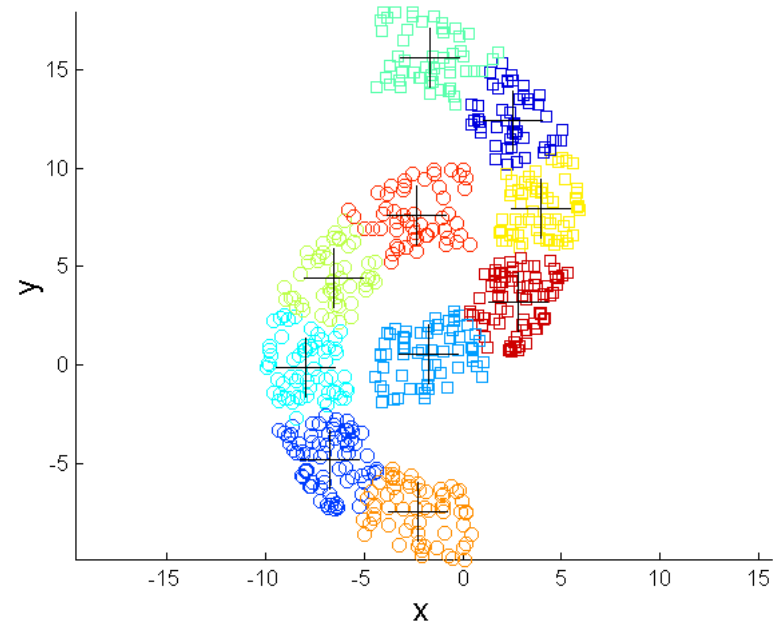


K-means (2 Clusters)

Overcoming K-means Limitations



Original Points

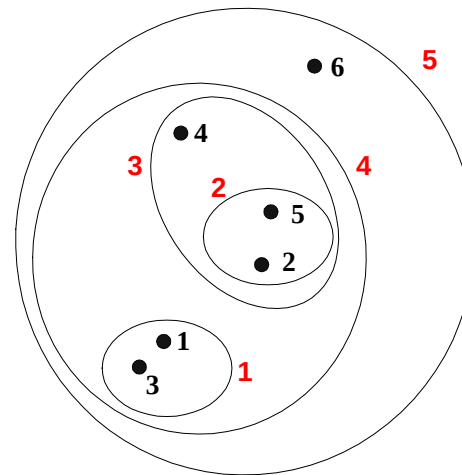
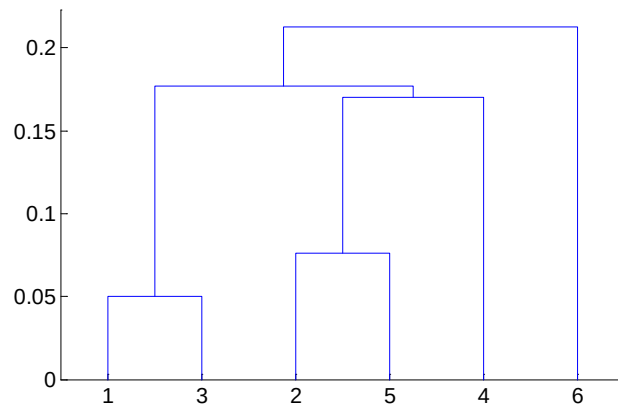


K-means Clusters

One solution is to find a large number of clusters such that each of them represents a part of a natural cluster. But these small clusters need to be put together in a post-processing step.

Hierarchical Clustering

- Produces a set of nested clusters organized as a hierarchical tree
- Can be visualized as a dendrogram
 - A tree like diagram that records the sequences of merges or splits



Strengths of Hierarchical Clustering

- Do not have to assume any particular number of clusters
 - Any desired number of clusters can be obtained by ‘cutting’ the dendrogram at the proper level
- They may correspond to meaningful taxonomies
 - Example in biological sciences (e.g., animal kingdom, phylogeny reconstruction, ...)

Hierarchical Clustering

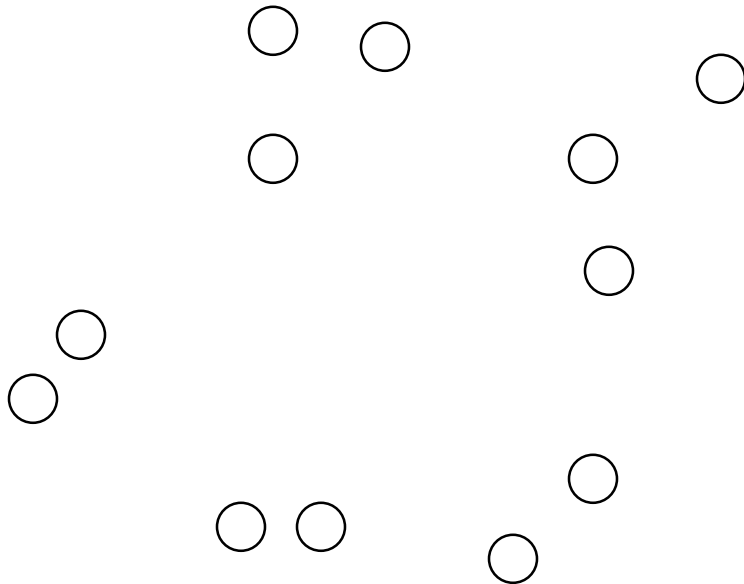
- Two main types of hierarchical clustering
 - Agglomerative:
 - ◆ Start with the points as individual clusters
 - ◆ At each step, merge the closest pair of clusters until only one cluster (or k clusters) left
 - Divisive:
 - ◆ Start with one, all-inclusive cluster
 - ◆ At each step, split a cluster until each cluster contains an individual point (or there are k clusters)
- Traditional hierarchical algorithms use a similarity or distance matrix
 - Merge or split one cluster at a time

Agglomerative Clustering Algorithm

- **Key Idea: Successively merge closest clusters**
- Basic algorithm
 1. Compute the proximity matrix
 2. Let each data point be a cluster
 3. **Repeat**
 4. Merge the two closest clusters
 5. Update the proximity matrix
 6. **Until** only a single cluster remains
- Key operation is **the computation of the proximity of two clusters**
 - Different approaches to defining the distance between clusters distinguish the different algorithms

Steps 1 and 2

- Start with clusters of individual points and a proximity matrix



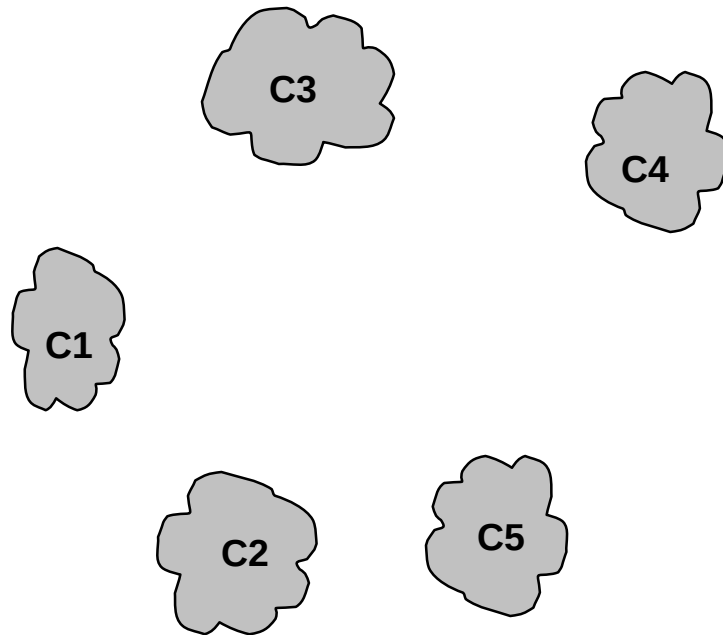
	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix



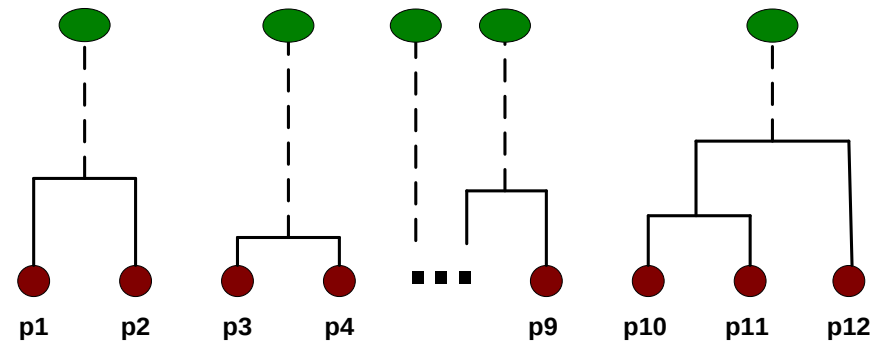
Intermediate Situation

- After some merging steps, we have some clusters



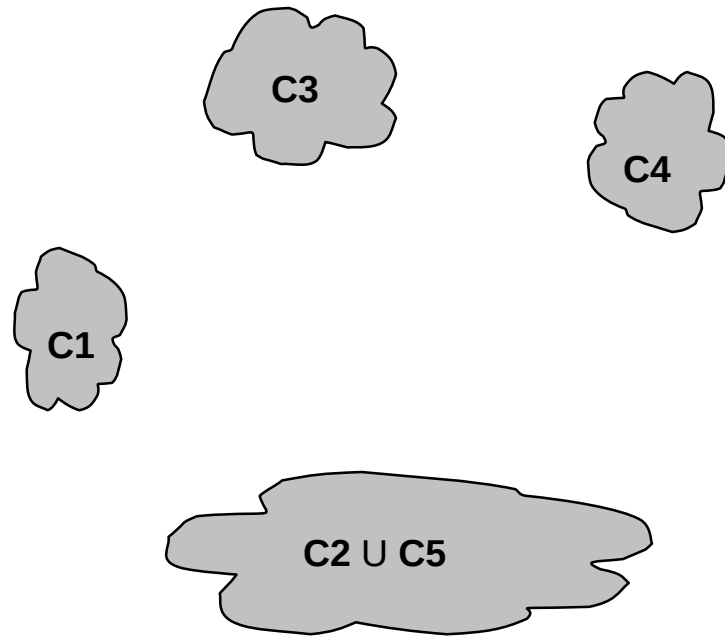
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix



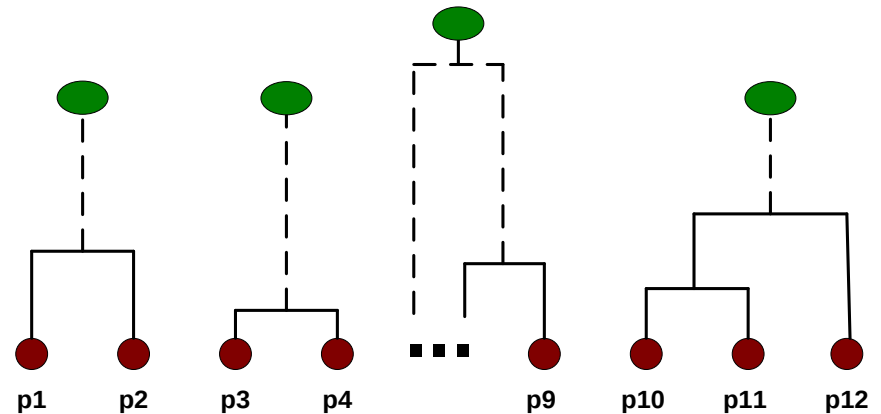
Step 5

- The question is “How do we update the proximity matrix?”

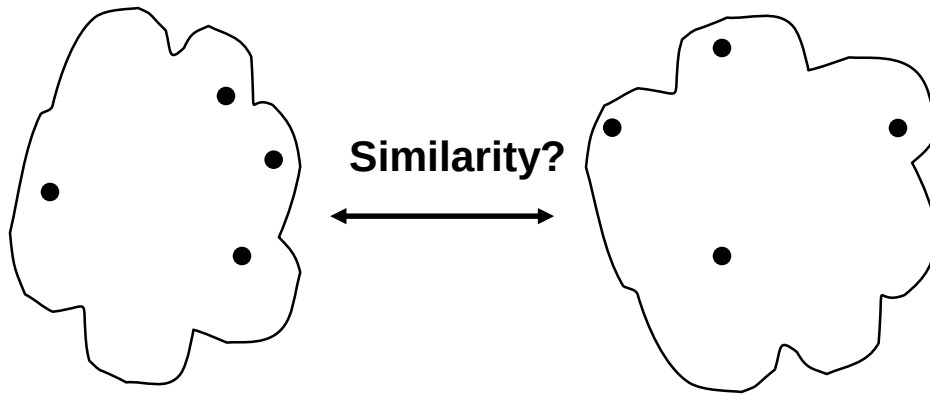


		$C2$ $\cup$ $C5$			
		C1	C5	C3	C4
C1			?		
$C2 \cup C5$		?	?	?	?
C3			?		
C4			?		

Proximity Matrix



How to Define Inter-Cluster Distance



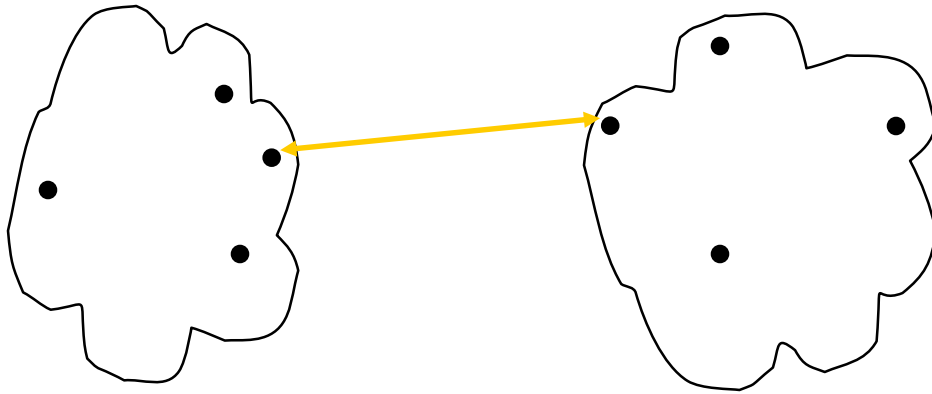
- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	.
p1						.
p2						
p3						
p4						
p5						
.						

	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix

How to Define Inter-Cluster Similarity

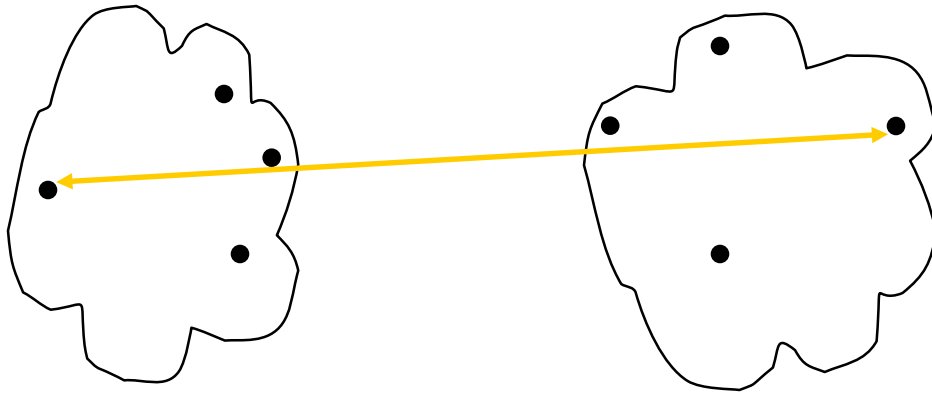


- **MIN (single link)**
 - the proximity between the closest two points that are in different clusters, or using graph terms, the shortest edge between two nodes in different subsets of nodes.

	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix

How to Define Inter-Cluster Similarity



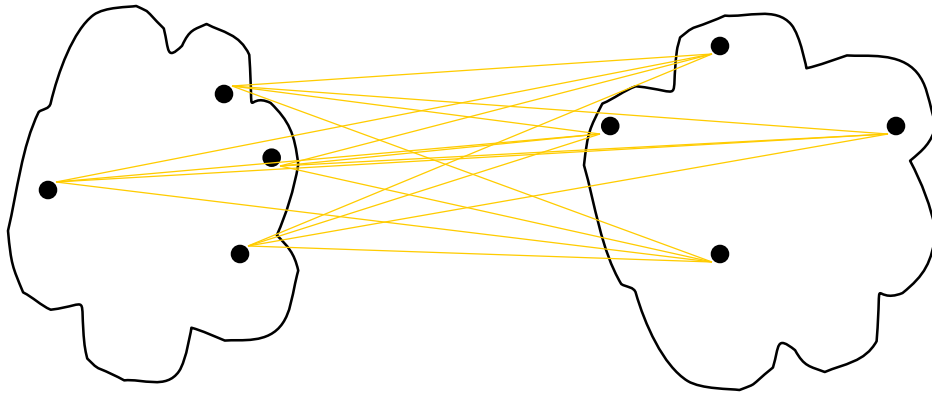
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

- **MAX (complete link)**

- The proximity between the farthest two points in different clusters, or using graph terms, the longest edge between two nodes in different subsets of nodes.

Proximity Matrix

How to Define Inter-Cluster Similarity



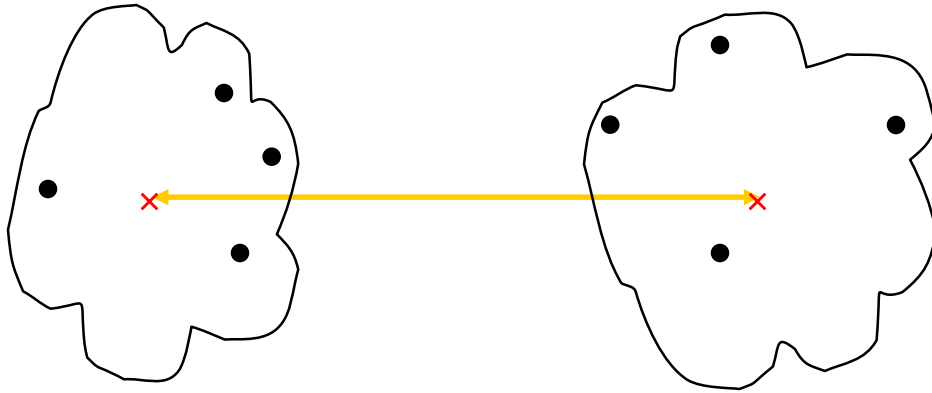
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

- **Group Average**

- The average pairwise proximities (average length of edges) of all pairs of points from different clusters.

Proximity Matrix

How to Define Inter-Cluster Similarity



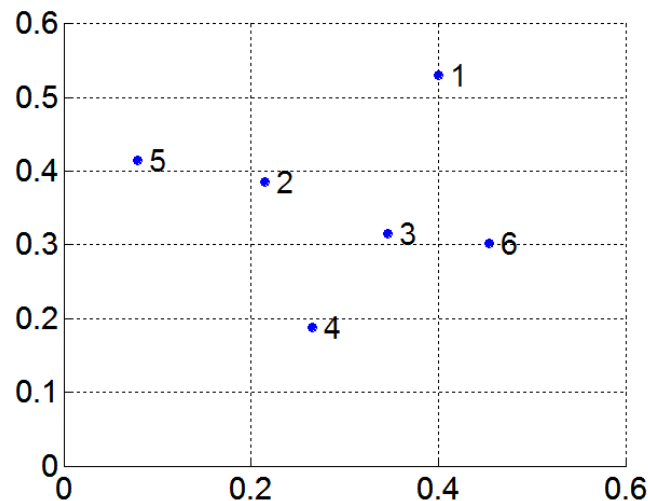
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix

- Distance Between Centroids
- Ward's Method uses squared error
 - The SSE change that results from merging the two clusters

MIN or Single Link

- Proximity of two clusters is based on the two closest points in the different clusters
 - Determined by one pair of points, i.e., by one link in the proximity graph
- Example:

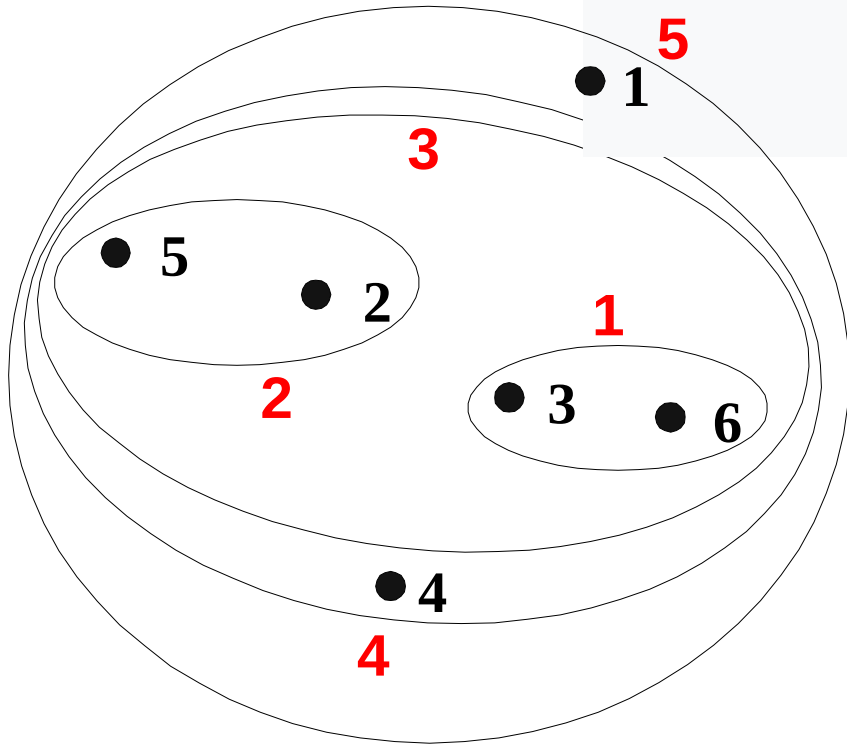


Distance Matrix:

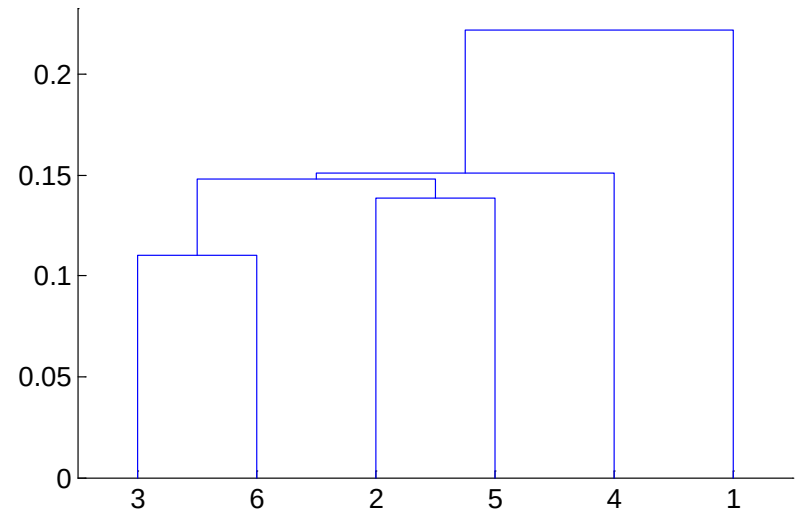
	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

Hierarchical Clustering: MIN

$$\begin{aligned} \text{dist}(\{3, 6\}, \{2, 5\}) &= \min(\text{dist}(3, 2), \text{dist}(6, 2), \text{dist}(3, 5), \text{dist}(6, 5)) \\ &= \min(0.15, 0.25, 0.28, 0.39) \\ &= 0.15. \end{aligned}$$



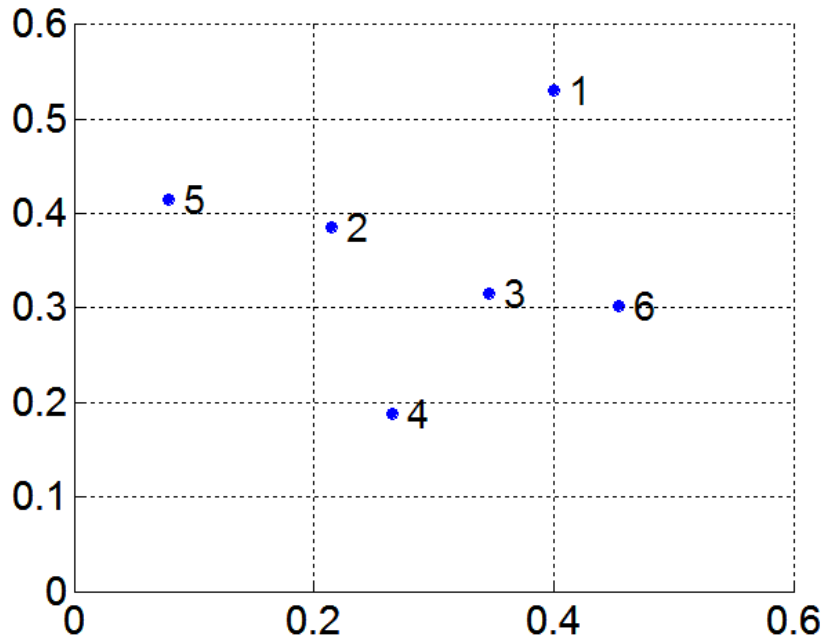
Nested Clusters



Dendrogram

MAX or Complete Linkage

- Proximity of two clusters is based on the two most distant points in the different clusters
 - Determined by all pairs of points in the two clusters



Distance Matrix:

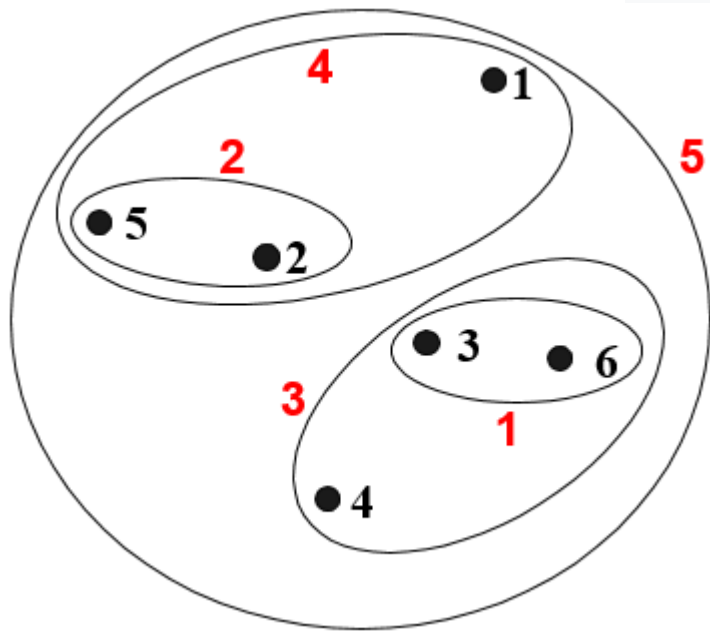
	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

Hierarchical Clustering: MAX

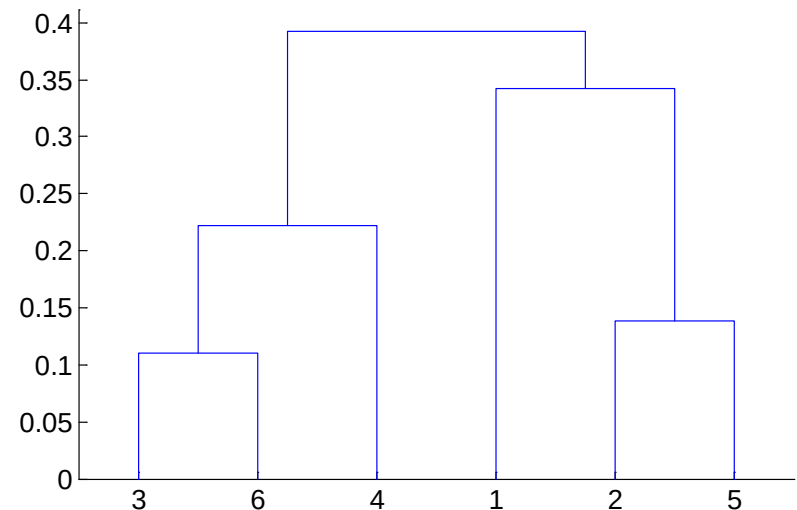
$$\begin{aligned} \text{dist}(\{3, 6\}, \{4\}) &= \max(\text{dist}(3, 4), \text{dist}(6, 4)) \\ &= \max(0.15, 0.22) \\ &= 0.22. \end{aligned}$$

$$\begin{aligned} \text{dist}(\{3, 6\}, \{2, 5\}) &= \max(\text{dist}(3, 2), \text{dist}(6, 2), \text{dist}(3, 5), \text{dist}(6, 5)) \\ &= \max(0.15, 0.25, 0.28, 0.39) \\ &= 0.39. \end{aligned}$$

$$\begin{aligned} \text{dist}(\{3, 6\}, \{1\}) &= \max(\text{dist}(3, 1), \text{dist}(6, 1)) \\ &= \max(0.22, 0.23) \\ &= 0.23. \end{aligned}$$



Nested Clusters

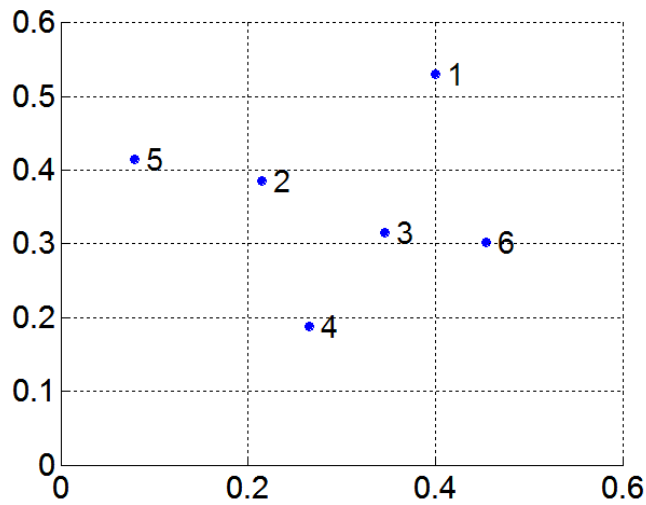


Dendrogram

Group Average

- Proximity of two clusters is the average of pairwise proximity between points in the two clusters.

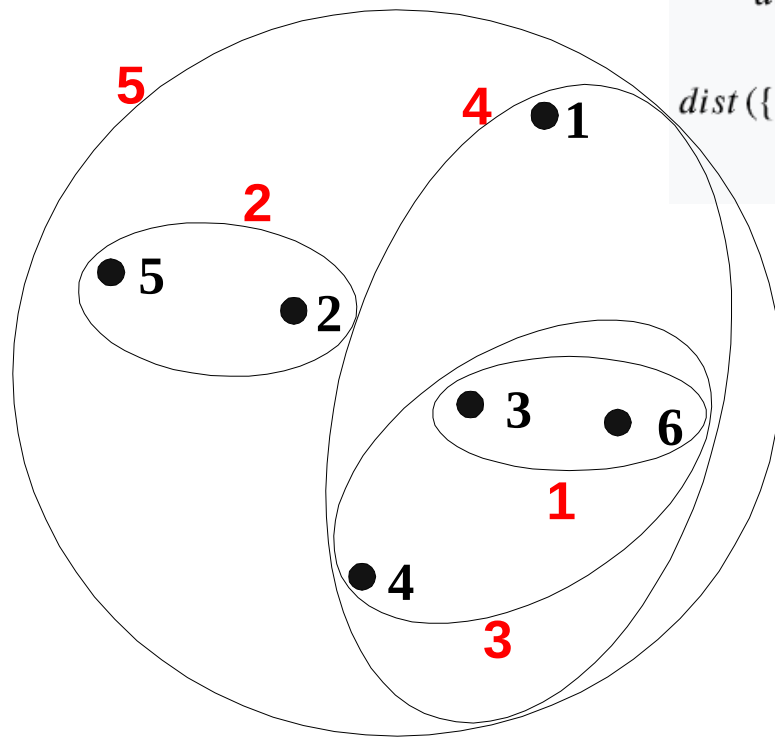
$$\text{proximity}(\text{Cluster}_i, \text{Cluster}_j) = \frac{\sum_{\substack{p_i \in \text{Cluster}_i \\ p_j \in \text{Cluster}_j}} \text{proximity}(p_i, p_j)}{|\text{Cluster}_i| \times |\text{Cluster}_j|}$$



Distance Matrix:

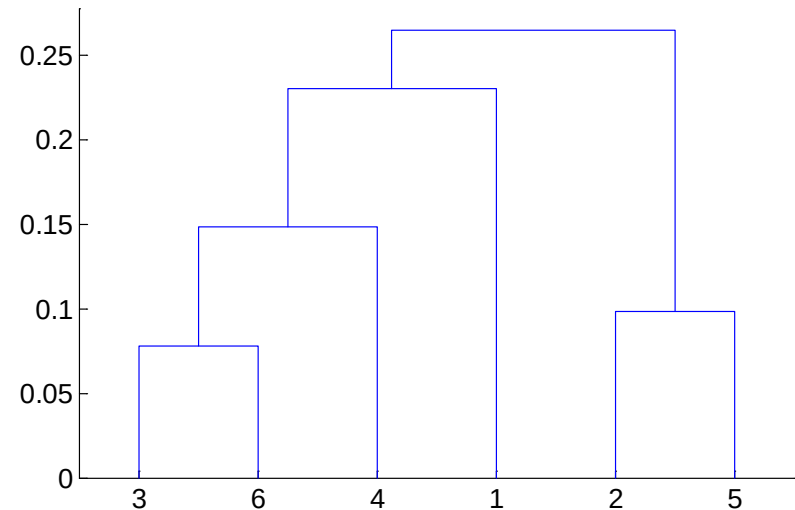
	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

Hierarchical Clustering: Group Average



Nested Clusters

$$\begin{aligned} \text{dist}(\{3, 6, 4\}, \{1\}) &= (0.22 + 0.37 + 0.23) / (3 \times 1) \\ &= 0.28 \\ \text{dist}(\{2, 5\}, \{1\}) &= (0.24 + 0.34) / (2 \times 1) \\ &= 0.29 \\ \text{dist}(\{3, 6, 4\}, \{2, 5\}) &= (0.15 + 0.28 + 0.25 + 0.39 + 0.20 + 0.29) / (3 \times 2) \\ &= 0.26 \end{aligned}$$



Dendrogram

Cluster Similarity: Ward's Method

- Similarity of two clusters is based on the increase in squared error when two clusters are merged
 - Similar to group average if distance between points is distance squared
- Less susceptible to noise
- Biased towards globular clusters
- Hierarchical analogue of K-means
 - Can be used to initialize K-means

Within-cluster variance

$$W(C) = \sum_{x \in C} \|x - \mu_C\|^2$$

Where μ_C is the centroid of cluster C
and $\|\cdot\|$ denotes the Euclidean distance.

Increase in variance

$$\Delta W = W(C_1) + W(C_2) - W(C_{12})$$

Where $W(C_{12})$ is the within-cluster variance of the new cluster formed by merging C_1 and C_2 .

● Merging Process

1. Calculate Within-Cluster Variance:

For all pairs of clusters, calculate their individual within-cluster variances $W(C1)$ and $W(C2)$.

2. Form New Cluster:

Compute the centroid of the new cluster $C12$ that would result from merging $C1$ and $C2$. The centroid is the mean of all points in both clusters.

3. Compute Increase in Variance:

Calculate the increase in total within-cluster variance ΔW for merging each pair of clusters.

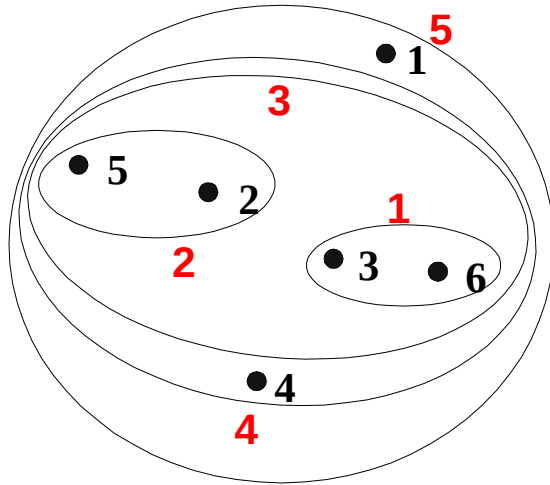
4. Choose the Pair with Minimum Increase:

Ward's method selects the pair of clusters whose merge results in the **smallest increase** in within-cluster variance. This leads to the most compact clusters being formed during the merging process.

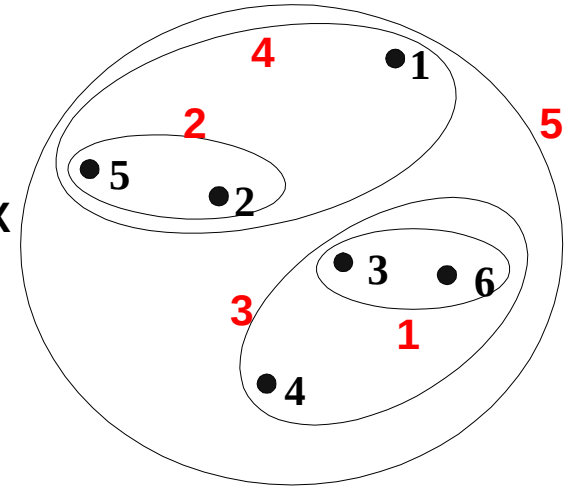
5. Repeat:

This process is repeated until all data points are merged into a single cluster or until a specified number of clusters is achieved.

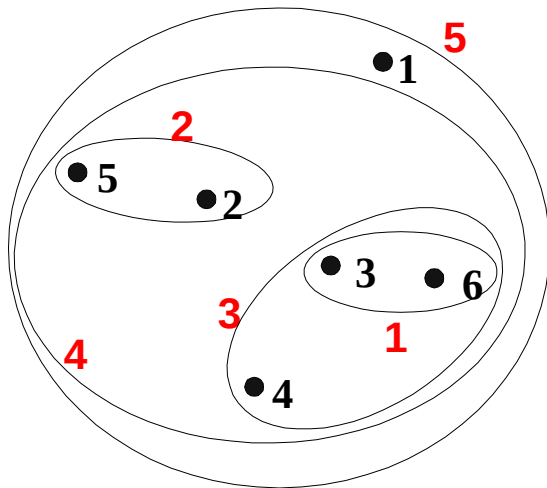
Hierarchical Clustering: Comparison



MIN

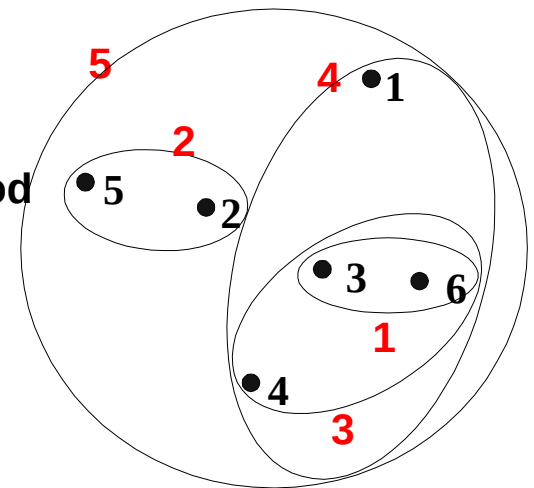


MAX



Group Average

Ward's Method



Hierarchical Clustering: Time and Space requirements

- $O(N^2)$ space since it uses the proximity matrix.
 - N is the number of points.
- $O(N^3)$ time in many cases
 - There are N steps and at each step the size, N^2 , proximity matrix must be updated and searched
 - Complexity can be reduced to $O(N^2 \log(N))$ time with some cleverness

Hierarchical Clustering: Problems and Limitations

- Once a decision is made to combine two clusters, it cannot be undone
- No global objective function is directly minimized
- Different schemes have problems with one or more of the following:
 - Sensitivity to noise
 - Difficulty handling clusters of different sizes and non-globular shapes
 - Breaking large clusters

Density Based Clustering

- Clusters are regions of high density that are separated from one another by regions of low density.

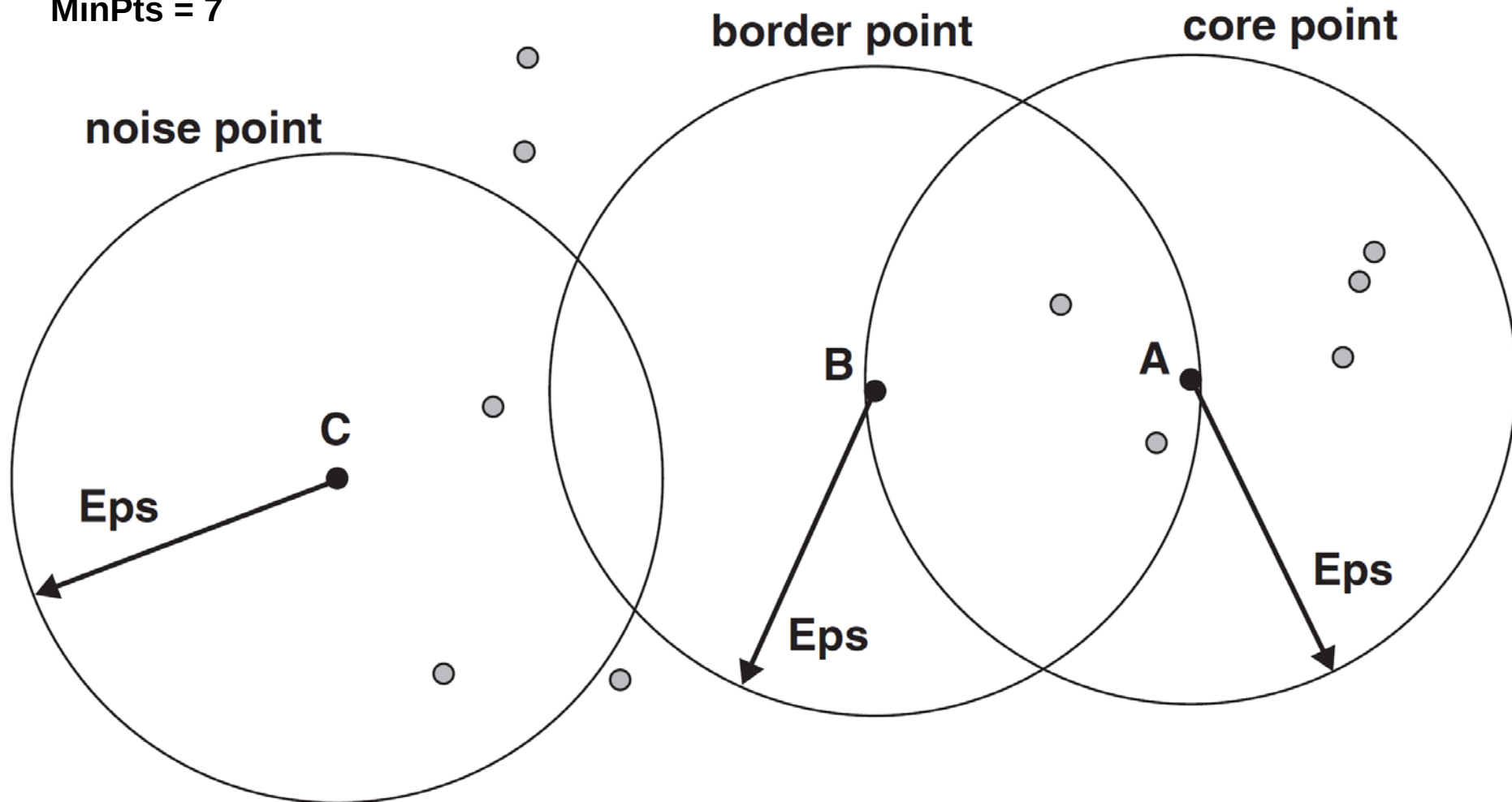


DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

- DBSCAN is a density-based algorithm.
 - Density = number of points within a specified radius (Eps)
 - A point is a **core point** if it has at least a specified number of points (MinPts) within Eps
 - ◆ These are points that are at the interior of a cluster
 - ◆ Counts the point itself
 - A **border point** is not a core point, but is in the neighborhood of a core point
 - A **noise point** is any point that is not a core point or a border point

DBSCAN: Core, Border, and Noise Points

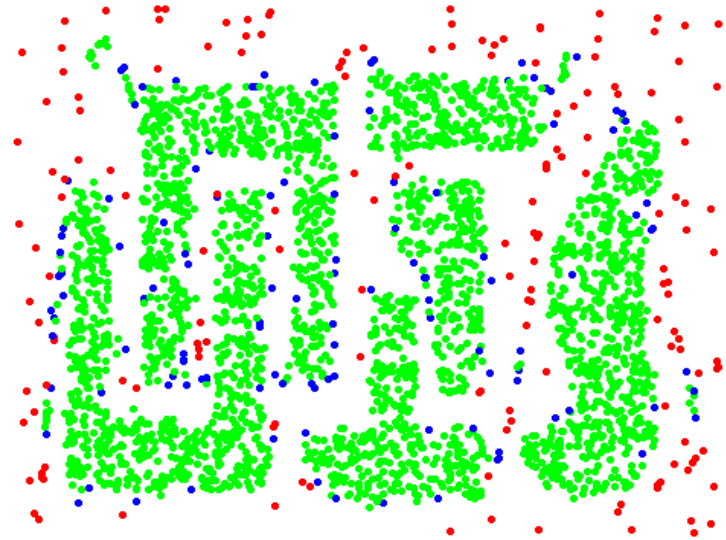
MinPts = 7



DBSCAN: Core, Border and Noise Points



Original Points



Point types: **core**,
border and **noise**

Eps = 10, MinPts = 4

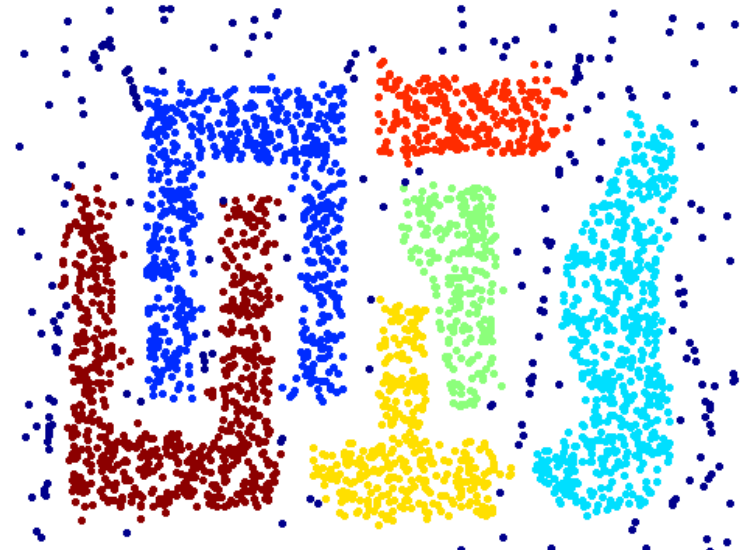
DBSCAN Algorithm

- Form clusters using core points, and assign border points to one of its neighboring clusters
- 1: Label all points as core, border, or noise points.
 - 2: Eliminate noise points.
 - 3: Put an edge between all core points within a distance Eps of each other.
 - 4: Make each group of connected core points into a separate cluster.
 - 5: Assign each border point to one of the clusters of its associated core points

When DBSCAN Works Well



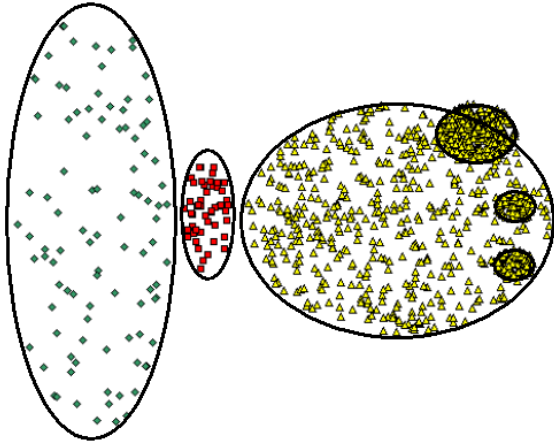
Original Points



Clusters (dark blue points indicate noise)

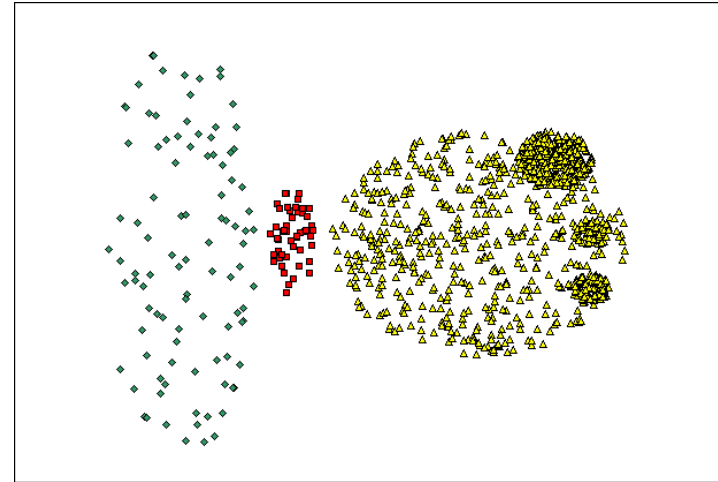
- Can handle clusters of different shapes and sizes
- Resistant to noise

When DBSCAN Does NOT Work Well

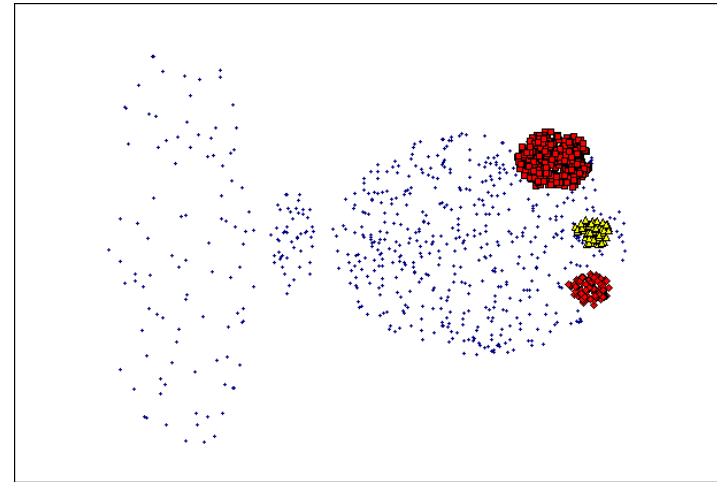


Original Points

- Varying densities
- High-dimensional data



(MinPts=4, Eps=9.92).



(MinPts=4, Eps=9.75)

DBSCAN: Determining EPS and MinPts

- Idea is that for points in a cluster, their k^{th} nearest neighbors are at close distance
- Noise points have the k^{th} nearest neighbor at farther distance
- So, plot sorted distance of every point to its k^{th} nearest neighbor

